

Race Performance Report

SESSION

Race

LAYOUT

Circuit de Spa-Francorchamps

CAR CLASS

Hyper

TEAM

-

DURATION

144:45.450

DISTANCE

459.1 km

COMPOUNDS

1.0

SOC START / END

100.0 / 49.6

GENERATED

2026-07-16T23:26+02:00

CIRCUIT

Circuit de Spa-Francorchamps

CAR

Toyota GR010 Custom Team 2024 #397

DRIVER

Driver unavailable

SESSION DATE

2026-05-16T19:48:11.980235

LAPS VALID / TOTAL

52 / 67

PIT ENTRIES / EXITS

2 / 2

FUEL START / END

85.0 / 20.1 L

FINISH / CLASSIFICATION

- / -

Executive Summary / Sintesi

RECORDED / VALID LAPS

67 / 52

BEST VALID LAP

2:03.248

REPRESENTATIVE PACE

2:05.006

CONSISTENCY MAD

0.630 s

BEST SUSTAINED PACE

2:04.494

BEST 3 / 5 LAP AVG

2:03.966 / 2:04.170

FUEL / LAP

3.527 L

STINTS / PIT STOPS

4 / 2

CONFIDENCE

High

Key findings

- Representative Pace: 2:05.006 (measured, high confidence). Median of 52 clean valid laps.
- Pace Trend: 0.021 s/lap (inferred, high confidence). Theil-Sen trend across clean laps.
- Fuel Effect: 0.001 s/L (inferred, medium confidence). robust pairwise estimate; fuel remains correlated with tyre age and track evolution
- Tyre Degradation: -0.027 s/lap (inferred, low confidence). inferred from within-stint clean-pace trend; fuel, traffic and track evolution cannot be fully separated

Position progression is shown only when measured; position changes are not called overtakes without opponent and pit-cycle evidence.

Data Quality & Methodology

52 laps included and 15 excluded. Missing channels: position. Overall confidence: High. Excluded laps remain in the audit.

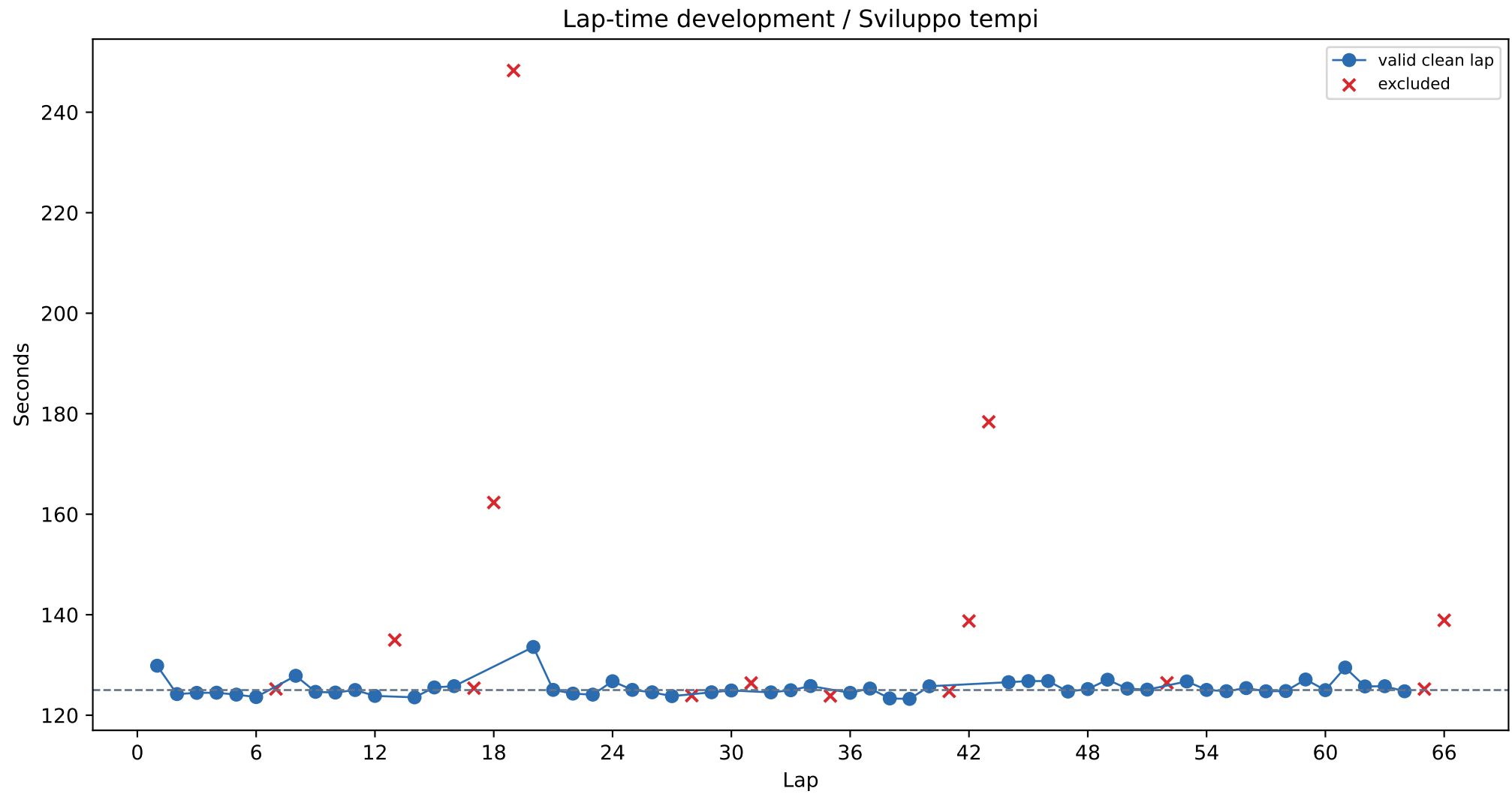
Duplicate samples	-
Timestamp discontinuities	-
Impossible samples	-
Samples analyzed	4992
Missing channels	position
Source warnings	GPS Time: sampled every 5 rows for review performance.

Filters

shared lap-quality contract - finite timing 40-900 s - pit/yellow/incomplete exclusion - sensor anomaly audit

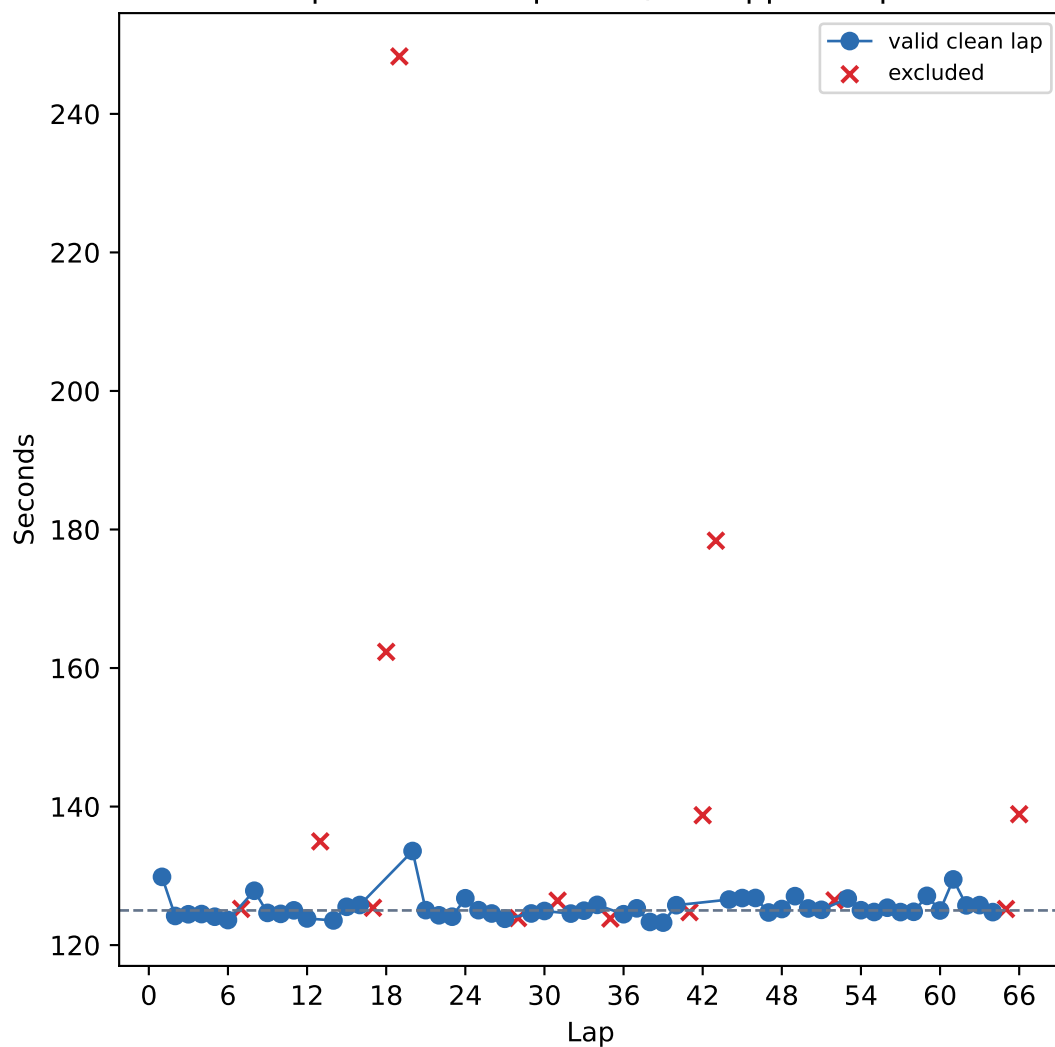
Methods: median, trimmed mean, MAD, Theil-Sen pairwise slope. Measured values and labeled inferences only; no external LLM.

Session Timeline

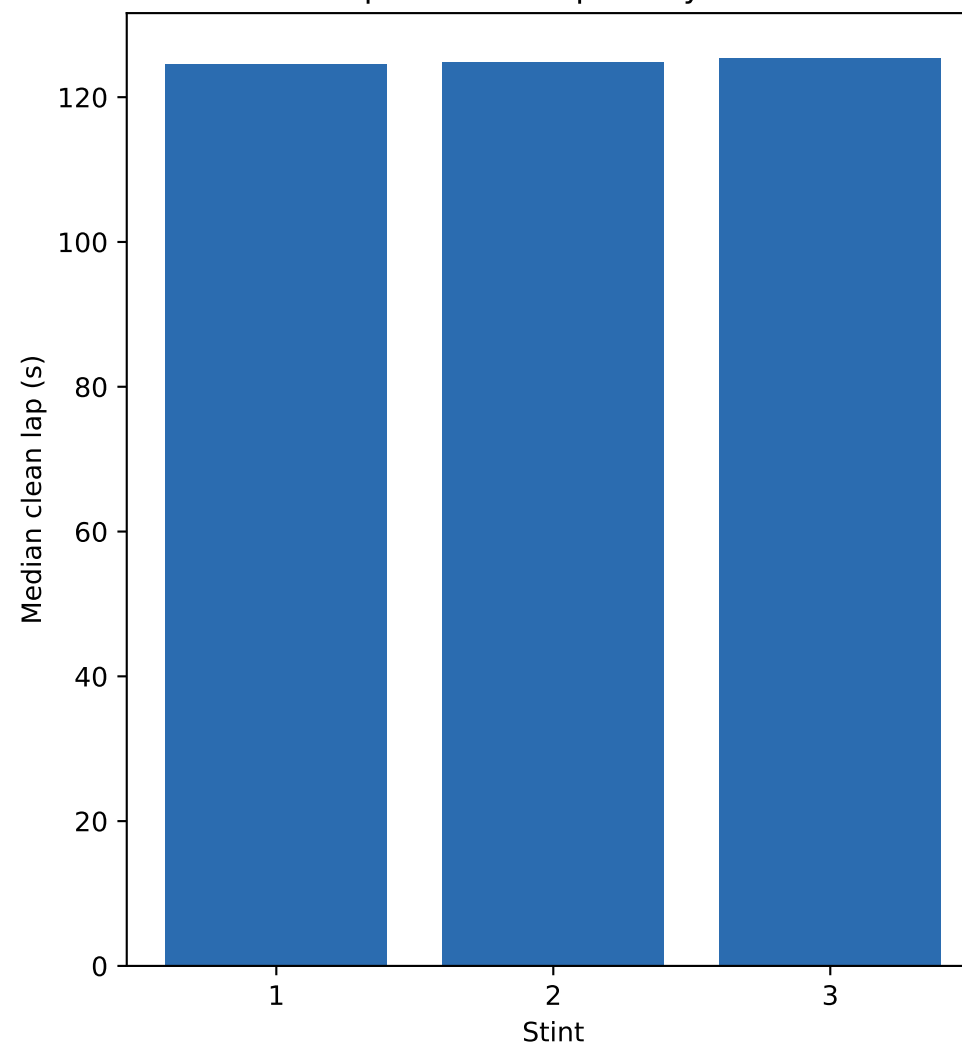


Lap & Stint Analysis

Lap-time development / Sviluppo tempi

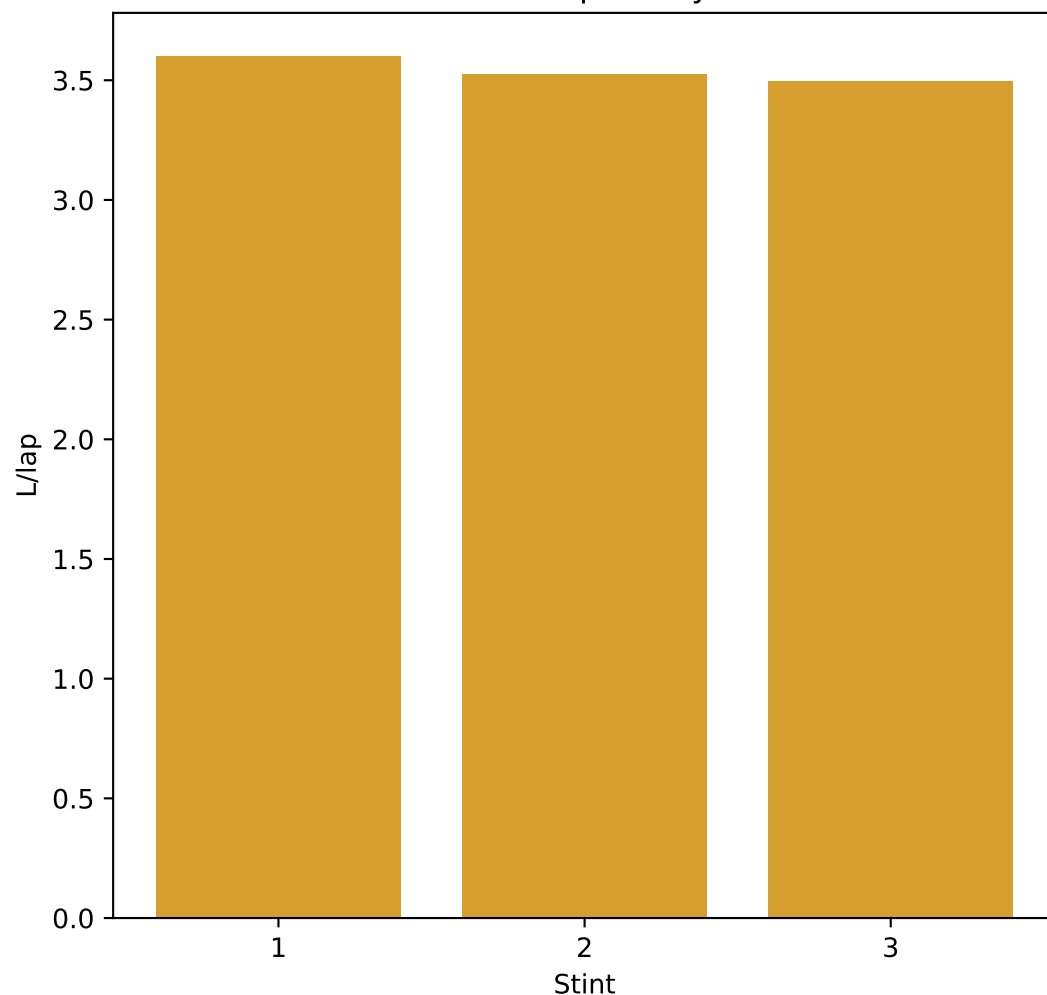


Representative pace by stint



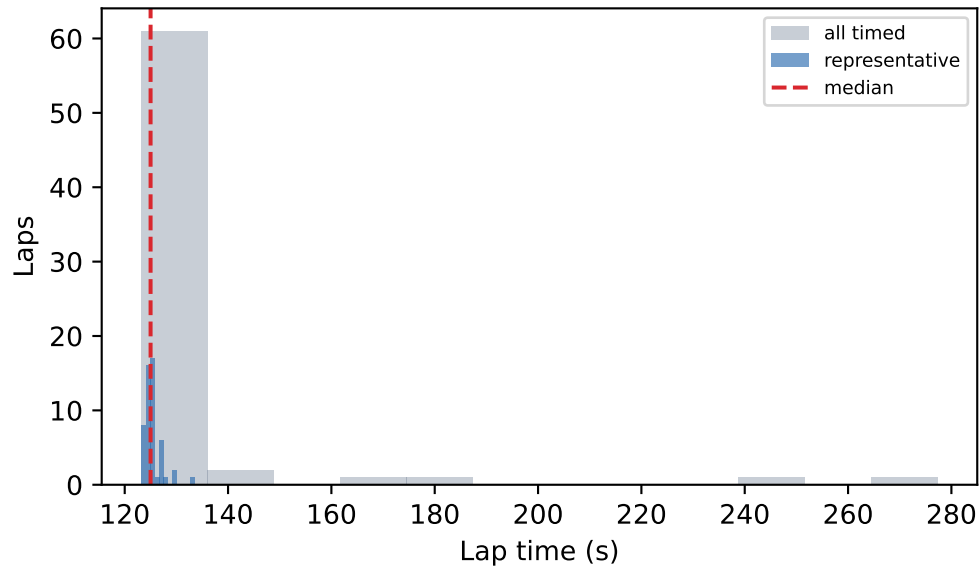
Fuel & Tyre Analysis

Fuel consumption by stint



Fuel median: 3.527 L/lap Fuel effect: 0.0010 s/L (medium) robust pairwise estimate; fuel remains correlated with tyre age and track evolution Tyre degradation: -0.027 s/lap (low) inferred from within-stint clean-pace trend; fuel, traffic and track evolution cannot be fully separated

Distribution, Energy, Thermal & Environment



Energy

SoC: 100.00 -> 49.59 Virtual energy: 100.00 -> 20.33 Observed states only; deployment targets require vehicle-specific configuration.

Thermal

Peak tyre: 108.8 C Peak brake: 701.1 C Oil max: 114.4 C; water max: 97.9 C Observed extremes; no ideal or warning threshold is claimed without vehicle-specific targets.

Environment

Track temp change: 3.6 C Ambient change: 4.0 C Wetness change: 0.00

Platform

Front ride height min: 0.016 Rear ride height min: 0.008 Absolute aero coefficients and bottoming thresholds are unavailable without vehicle definitions.

Robust distribution metrics

Mean (trimmed): 2:05.168 Median: 2:05.006 Std dev: 1.757 s IQR: 1.277 s MAD: 0.630 s

Best consecutive 3-lap average: 2:03.966 Best consecutive 5-lap average: 2:04.170

Best Lap, Potential & Comparison

BEST ACTUAL

2:03.248

THEORETICAL BEST

-

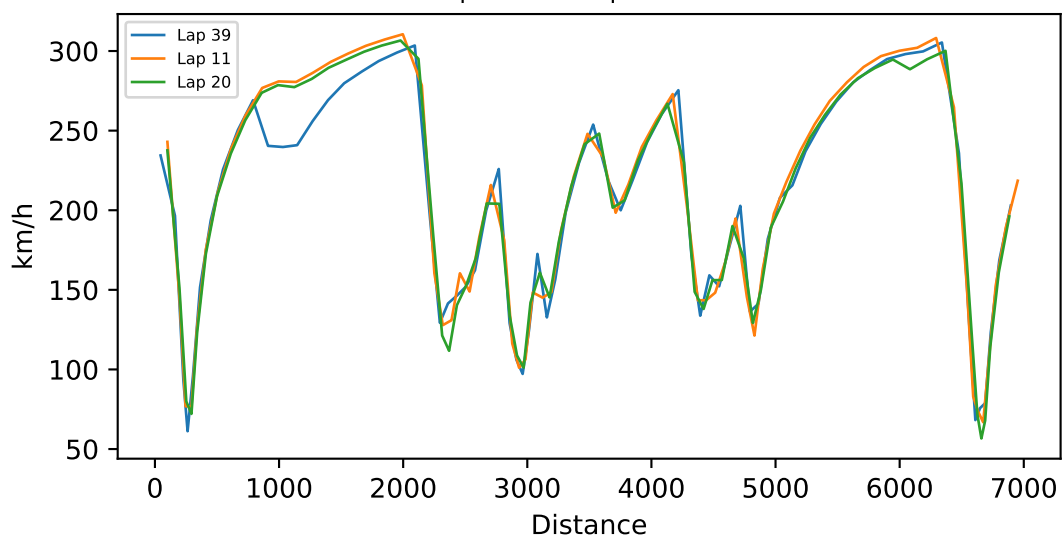
REALISTIC POTENTIAL

2:03.248

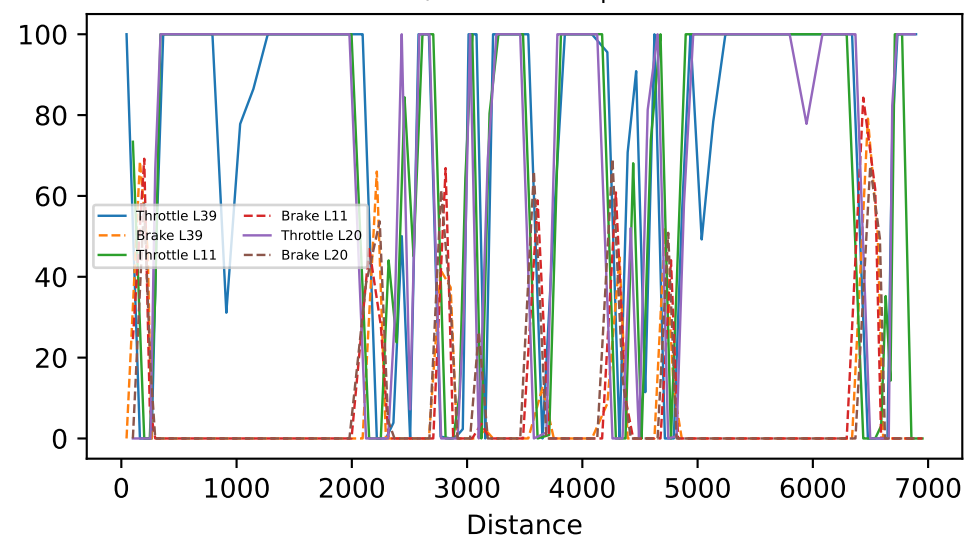
GAP

-

Speed over lap distance



Throttle / brake over lap distance



Corner analysis

no shared corner map and compatible reference-lap segmentation are available

The theoretical best is not presented as automatically achievable. A realistic composite is only produced when compatible sector evidence exists.

Pits, Traffic & Event Evidence

Pit stops

Detected: 2. opponent pit-cycle state is not recorded reliably

Traffic

opponent gaps, identifiers and reliable position data are unavailable

Position

position channel unavailable

Corners

no shared corner map and compatible reference-lap segmentation are available

Race Performance Assessment

Position progression is shown only when measured; position changes are not called overtakes without opponent and pit-cycle evidence.

Race Time	144:45.450
Completed / Valid Laps	67 / 52
Distance	459.1 km
Best Race Lap	2:03.248
Median Clean Pace	2:05.006
Best Stint	1
Pit Stops	2
Fuel Used	183.3 L

Complete Lap Table - laps 1 to 23

Every recorded lap is retained. A dash means the source channel or reliable derivation is unavailable.

Lap	Time	dBest	dMed	Classification	St	Comp	Fuel S	Fuel E	Used	SoC S	SoC E	Energy	Vmax	Tyre C	Brake C	Traffic	Off/Imp
1	2:09.851	6.60	4.85	valid representative	1	1	81.2	77.9	3.31	49.31	14.22	4.06	310	76	600	unknown	0/1.0
2	2:04.208	0.96	-0.80	valid representative	1	1	77.9	74.3	3.57	11.62	11.47	4.16	309	79	642	unknown	0/-
3	2:04.450	1.20	-0.56	valid representative	1	1	74.3	70.8	3.53	9.32	10.13	4.17	305	83	634	unknown	0/-
4	2:04.480	1.23	-0.53	valid representative	1	1	70.8	67.2	3.60	8.20	11.40	4.15	308	84	647	unknown	0/-
5	2:04.098	0.85	-0.91	valid representative	1	1	67.2	63.6	3.62	9.26	11.39	4.14	309	86	647	unknown	0/0.0
6	2:03.615	0.37	-1.39	valid representative	1	1	63.6	60.0	3.56	9.31	11.96	4.10	309	86	652	unknown	0/-
7	2:05.227	1.98	0.22	excluded, off_track_surface,	1	1	60.0	56.6	3.43	9.72	13.06	4.06	302	82	591	unknown	1/1.0
8	2:07.847	4.60	2.84	valid representative	1	1	56.6	52.9	3.67	12.52	42.21	4.01	306	85	600	unknown	0/-
9	2:04.648	1.40	-0.36	valid representative	1	1	52.9	49.3	3.66	41.59	61.99	4.16	309	86	619	unknown	0/-
10	2:04.508	1.26	-0.50	valid representative	1	1	49.3	45.6	3.65	60.74	67.12	4.20	307	85	656	unknown	0/-
11	2:05.017	1.77	0.01	valid representative	1	1	45.6	42.1	3.49	65.21	55.39	4.08	311	86	634	unknown	0/-
12	2:03.846	0.60	-1.16	valid representative	1	1	42.1	38.5	3.62	53.51	46.43	4.07	310	86	661	unknown	0/-
13	2:14.972	11.72	9.97	excluded, off_track_surface,	1	1	38.5	35.0	3.52	44.56	32.81	4.15	309	87	639	unknown	1/-
14	2:03.559	0.31	-1.45	valid representative	1	1	35.0	31.4	3.59	31.60	46.11	4.07	309	85	635	unknown	0/-
15	2:05.530	2.28	0.52	valid representative	1	1	31.4	27.8	3.61	44.86	58.98	4.17	304	86	641	unknown	0/-
16	2:05.789	2.54	0.78	valid representative	1	1	27.8	24.2	3.61	57.77	76.77	4.08	308	84	602	unknown	0/-
17	2:05.386	2.14	0.38	excluded, pit_lap, off_track	-	1	24.2	21.1	3.10	75.62	96.01	3.36	300	82	539	unknown	7/-
18	2:42.333	39.09	37.33	excluded, lap_time_outside_s	-	1	21.1	81.9	-	96.07	95.33	-71.82	276	69	566	unknown	0/-
19	4:08.296	125.05	123.29	excluded, lap_time_outside_s	2	1	81.9	78.5	3.40	94.08	93.26	3.86	300	63	627	clear	3/-
20	2:13.584	10.34	8.58	valid representative	2	1	78.5	75.0	3.53	92.13	91.58	3.93	307	71	685	clear	0/-
21	2:05.045	1.80	0.04	valid representative	2	1	75.0	71.4	3.61	90.33	90.54	4.16	307	77	660	clear	0/-
22	2:04.302	1.05	-0.70	valid representative	2	1	71.4	67.8	3.62	89.30	91.80	4.07	309	81	689	unknown	0/-
23	2:04.100	0.85	-0.91	valid representative	2	1	67.8	64.3	3.51	90.56	92.02	4.00	302	81	676	unknown	0/-

Complete Lap Table - laps 24 to 46

Every recorded lap is retained. A dash means the source channel or reliable derivation is unavailable.

Lap	Time	dBest	dMed	Classification	St	Comp	Fuel S	Fuel E	Used	SoC S	SoC E	Energy	Vmax	Tyre C	Brake C	Traffic	Off/Imp
24	2:06.763	3.52	1.76	valid representative	2	1	64.3	60.7	3.57	90.77	90.00	4.06	309	82	652	unknown	0/-
25	2:05.043	1.80	0.04	valid representative	2	1	60.7	57.3	3.41	87.50	59.61	4.05	310	85	701	unknown	0/-
26	2:04.562	1.31	-0.44	valid representative	2	1	57.3	53.9	3.40	57.19	25.88	4.18	309	85	630	unknown	0/0.0
27	2:03.801	0.55	-1.20	valid representative	2	1	53.9	50.4	3.52	23.34	19.16	4.11	308	85	607	unknown	0/-
28	2:03.900	0.65	-1.11	excluded, off_track_surface,	2	1	50.4	46.9	3.50	17.29	18.77	3.98	310	85	634	unknown	1/-
29	2:04.566	1.32	-0.44	valid representative	2	1	46.9	43.4	3.48	16.90	18.08	4.03	309	84	586	unknown	0/-
30	2:04.925	1.68	-0.08	valid representative	2	1	43.4	39.9	3.53	16.21	20.35	4.09	307	84	618	unknown	0/-
31	2:06.418	3.17	1.41	excluded, off_track_surface,	2	1	39.9	36.3	3.55	18.48	19.44	4.07	310	84	619	unknown	1/-
32	2:04.548	1.30	-0.46	valid representative	2	1	36.3	32.8	3.54	17.62	18.32	4.09	307	82	644	unknown	0/-
33	2:04.971	1.72	-0.03	valid representative	2	1	32.7	29.3	3.47	17.07	33.16	4.01	307	83	628	unknown	0/-
34	2:05.809	2.56	0.80	valid representative	2	1	29.3	25.7	3.59	31.93	45.15	4.03	308	85	589	unknown	0/-
35	2:03.835	0.59	-1.17	excluded, off_track_surface,	2	1	25.7	22.1	3.56	43.89	56.39	3.97	311	85	616	unknown	1/-
36	2:04.458	1.21	-0.55	valid representative	2	1	22.1	18.5	3.58	55.14	70.56	4.05	308	83	588	unknown	0/-
37	2:05.309	2.06	0.30	valid representative	2	1	18.5	15.1	3.41	69.32	41.13	4.10	308	84	616	unknown	0/-
38	2:03.341	0.09	-1.66	valid representative	2	1	15.1	11.6	3.53	38.71	26.90	4.14	310	86	635	unknown	0/-
39	2:03.248	0.00	-1.76	valid representative	2	1	11.6	8.1	3.45	25.03	19.96	3.99	305	82	620	unknown	0/-
40	2:05.761	2.51	0.76	valid representative	2	1	8.1	4.7	3.48	18.09	16.58	4.09	310	83	605	unknown	0/-
41	2:04.758	1.51	-0.25	excluded, pit_lap, pit	-	1	4.6	1.3	3.36	14.76	25.02	3.78	311	84	616	unknown	0/-
42	2:18.751	15.50	13.75	excluded, pit_lap, pit	-	1	1.3	81.6	-	25.02	22.70	-94.28	305	78	592	unknown	0/-
43	2:58.382	55.13	53.38	excluded, lap_time_outside_s	3	1	81.6	78.1	3.48	20.83	22.01	4.08	306	81	615	unknown	0/-
44	2:06.577	3.33	1.57	valid representative	3	1	78.1	74.5	3.60	20.19	35.44	4.13	306	81	637	unknown	0/1.0
45	2:06.791	3.54	1.79	valid representative	3	1	74.5	70.9	3.59	34.19	51.22	4.10	309	79	642	unknown	0/0.0
46	2:06.804	3.56	1.80	valid representative	3	1	70.9	67.3	3.63	49.97	63.66	4.06	310	84	621	unknown	0/-

Complete Lap Table - laps 47 to

Every recorded lap is retained. A dash means the source channel or reliable derivation is unavailable.

Lap	Time	dBest	dMed	Classification	St	Comp	Fuel S	Fuel E	Used	SoC S	SoC E	Energy	Vmax	Tyre C	Brake C	Traffic	Off/Imp
47	2:04.708	1.46	-0.30	valid representative	3	1	67.3	63.7	3.57	62.42	63.13	4.10	308	85	618	unknown	0/-
48	2:05.208	1.96	0.20	valid representative	3	1	63.7	60.2	3.52	61.25	59.30	4.09	308	85	618	unknown	0/-
49	2:07.066	3.82	2.06	valid representative	3	1	60.2	56.8	3.39	57.43	50.49	4.01	309	86	606	unknown	0/-
50	2:05.290	2.04	0.28	valid representative	3	1	56.8	53.3	3.52	48.62	40.52	4.11	310	84	605	unknown	0/-
51	2:05.092	1.84	0.09	valid representative	3	1	53.3	49.9	3.37	38.18	34.89	4.00	309	86	625	unknown	0/-
52	2:06.450	3.20	1.44	excluded, off_track_surface,	3	1	49.9	46.4	3.52	33.02	29.71	4.06	308	84	598	unknown	1/-
53	2:06.725	3.48	1.72	valid representative	3	1	46.4	42.8	3.58	27.84	35.70	4.08	310	83	613	unknown	0/-
54	2:05.037	1.79	0.03	valid representative	3	1	42.8	39.2	3.62	34.45	46.49	4.02	309	85	616	unknown	0/-
55	2:04.779	1.53	-0.23	valid representative	3	1	39.2	35.7	3.52	45.28	39.72	4.09	309	85	614	unknown	0/-
56	2:05.398	2.15	0.39	valid representative	3	1	35.7	32.2	3.48	37.85	28.99	4.10	309	86	619	unknown	0/-
57	2:04.767	1.52	-0.24	valid representative	3	1	32.2	28.8	3.45	27.12	21.93	4.07	309	87	595	unknown	0/-
58	2:04.809	1.56	-0.20	valid representative	3	1	28.8	25.3	3.41	20.03	18.86	3.98	303	81	605	unknown	0/-
59	2:07.109	3.86	2.10	valid representative	3	1	25.3	21.9	3.49	16.99	16.53	4.05	310	84	621	unknown	0/-
60	2:04.994	1.75	-0.01	valid representative	3	1	21.9	18.4	3.42	14.66	38.85	3.81	311	109	617	unknown	0/-
61	2:09.484	6.24	4.48	valid representative	3	1	18.4	14.9	3.49	37.63	55.55	3.96	308	81	572	unknown	0/-
62	2:05.732	2.48	0.73	valid representative	3	1	14.9	11.5	3.41	53.67	46.42	4.03	308	81	598	unknown	0/-
63	2:05.774	2.53	0.77	valid representative	3	1	11.5	8.0	3.50	44.61	38.30	3.97	311	85	596	unknown	0/-
64	2:04.764	1.52	-0.24	valid representative	3	1	8.0	4.5	3.50	36.43	29.06	4.05	310	86	609	unknown	0/-
65	2:05.207	1.96	0.20	excluded, pit_lap, pit	-	1	4.5	1.1	3.40	27.81	49.46	3.87	306	85	631	unknown	0/-
66	2:18.896	15.65	13.89	excluded, pit_lap, pit	-	1	1.1	24.7	-	49.46	49.59	-19.07	60	65	350	unknown	0/-
-	4:37.300	154.05	152.29	excluded, lap_time_outside_s	-	-	85.0	81.2	3.79	-	-	-	305	-	-	-	0/-

Appendix A - Channel Availability

Coverage reflects the indexed source manifest. Metrics use full-resolution source calculations where available; report traces may be downsampled.

Channel	Hz	Rows	Unit	Mapped fields	Usage	Coverage
ABS	-	1	-	abs_active	direct/derived	indexed
ABSLevel	-	1	-	abs_level	direct/derived	indexed
Ambient Temperature	1	8686	C	ambient_temp	direct/derived	indexed
AntiStall Activated	-	1	-	anti_stall_active	direct/derived	indexed
Best LapTime	-	8	s	best_lap_time	direct/derived	indexed
Best Sector1	-	8	s	best_sector1	direct/derived	indexed
Best Sector2	-	8	s	best_sector2	direct/derived	indexed
Brake Bias Rear	-	1	-	brake_bias_rear	direct/derived	indexed
Brake Migration	-	1	-	brake_migration	direct/derived	indexed
Brake Pos	50	434274	%	brake	direct/derived	indexed
Brake Pos Unfiltered	50	434274	%	-	not used	indexed
Brake Thickness	10	86855	%	brake_thickness	direct/derived	indexed
Brakes Air Temp	50	434274	C	brake_air_temp	direct/derived	indexed
Brakes Force	50	434274	%	brake_force	direct/derived	indexed
Brakes Temp	50	434274	C	brake_temp	direct/derived	indexed
CloudDarkness	-	1	%	cloud_darkness	direct/derived	indexed
Clutch Pos	50	434274	%	clutch	direct/derived	indexed
Clutch Pos Unfiltered	50	434274	%	clutch_unfiltered	direct/derived	indexed
Clutch RPM	100	868548	RPM	clutch_rpm	direct/derived	indexed
Current LapTime	-	66	s	current_lap_time	direct/derived	indexed
Current Sector	-	201	-	sector	direct/derived	indexed
Current Sector1	-	67	s	sector1	direct/derived	indexed
Current Sector2	-	66	s	sector2	direct/derived	indexed
Engine Max RPM	-	1	RPM	-	not used	indexed
Engine Oil Temp	7	60951	C	engine_oil_temp	direct/derived	indexed
Engine RPM	100	868548	RPM	rpm	direct/derived	indexed
Engine Water Temp	7	60951	C	engine_water_temp	direct/derived	indexed
FFB Output	100	868548	%	ffb_output	direct/derived	indexed
Finish Status	-	1	-	finish_status	direct/derived	indexed
Front3rdDeflection	100	868548	m	front_3rd_deflection	direct/derived	indexed
FrontFlapActivated	-	1	-	front_flap_active	direct/derived	indexed
FrontRideHeight	100	868548	m	front_ride_height	direct/derived	indexed
Fuel Level	20	173710	L	fuel_liters	direct/derived	indexed
FuelMixtureMap	-	35	-	fuel_mixture_map	direct/derived	indexed
G Force Lat	10	86855	G	g_force_lat	direct/derived	indexed
G Force Long	10	86855	G	g_force_long	direct/derived	indexed
G Force Vert	10	86855	G	g_force_vert	direct/derived	indexed
GPS Latitude	10	86855	deg	gps_latitude	direct/derived	indexed
GPS Longitude	10	86855	deg	gps_longitude	direct/derived	indexed
GPS Speed	10	86855	m/s	-	not used	indexed
GPS Time	100	868548	s	game_time	direct/derived	indexed
Gear	-	6574	-	gear	direct/derived	indexed
Ground Speed	100	868548	km/h	speed_kph	direct/derived	indexed
Headlights State	-	201	On/Off	headlights	direct/derived	indexed
In Pits	-	6	-	in_pits	direct/derived	indexed
Lap	-	67	-	lap_number	direct/derived	indexed
Lap Dist	10	86855	m	lap_distance	direct/derived	indexed
Lap Time	-	67	s	last_lap_time	direct/derived	indexed
Last Sector1	-	67	s	last_sector1	direct/derived	indexed
Last Sector2	-	67	s	last_sector2	direct/derived	indexed
LastImpactMagnitude	-	43	-	last_impact_magnitude	direct/derived	indexed

Appendix A - Channel Availability

Coverage reflects the indexed source manifest. Metrics use full-resolution source calculations where available; report traces may be downsampled.

Channel	Hz	Rows	Unit	Mapped fields	Usage	Coverage
LaunchControlActive	-	1	-	-	not used	indexed
Minimum Path Wetness	-	1	%	minimum_path_wetness	direct/derived	indexed
OffpathWetness	-	1	%	offpath_wetness	direct/derived	indexed
OverheatingState	2	17371	-	overheating_state	direct/derived	indexed
Path Lateral	10	86855	m	path_lateral	direct/derived	indexed
Rear3rdDeflection	100	868548	m	rear_3rd_deflection	direct/derived	indexed
RearFlapActivated	-	1	-	rear_flap_active	direct/derived	indexed
RearFlapLegalStatus	-	1	-	rear_flap_legal	direct/derived	indexed
RearRideHeight	100	868548	m	rear_ride_height	direct/derived	indexed
Regen Rate	100	868548	kW	regen_rate	direct/derived	indexed
RideHeights	100	868548	m	ride_height	direct/derived	indexed
Sector1 Flag	-	65	-	sector1_flag	direct/derived	indexed
Sector2 Flag	-	47	-	sector2_flag	direct/derived	indexed
Sector3 Flag	-	31	-	sector3_flag	direct/derived	indexed
SoC	20	173710	%	soc	direct/derived	indexed
Speed Limiter	-	10	-	speed_limiter	direct/derived	indexed
Steering Pos	100	868548	%	steering	direct/derived	indexed
Steering Pos Unfiltered	100	868548	%	-	not used	indexed
Steering Shaft Torque	100	868548	Nm	steering_shaft_torque	direct/derived	indexed
SurfaceTypes	-	59919	-	surface_type	direct/derived	indexed
Susp Pos	100	868548	m	suspension_position	direct/derived	indexed
TC	-	52447	-	tc_active	direct/derived	indexed
TCCut	-	1	-	tc_cut	direct/derived	indexed
TCLLevel	-	1	-	tc_level	direct/derived	indexed
TCSlipAngle	-	1	-	tc_slip_angle	direct/derived	indexed
Throttle Pos	50	434274	%	throttle	direct/derived	indexed
Throttle Pos Unfiltered	50	434274	%	-	not used	indexed
Time Behind Next	2	17371	s	time_behind_next	direct/derived	indexed
Total Dist	10	86855	m	total_distance	direct/derived	indexed
Track Edge	10	86855	m	track_edge	direct/derived	indexed
Track Temperature	1	8686	C	track_temp	direct/derived	indexed
Turbo Boost Pressure	100	868548	Pa	turbo_boost_pressure	direct/derived	indexed
Tyres Wear	10	86855	%	tyre_wear	direct/derived	indexed
TyresCarcassTemp	5	43428	C	tyre_temp_carcass	direct/derived	indexed
TyresCompound	-	1	-	tyre_compound	direct/derived	indexed
TyresPressure	10	86855	kPa	tyre_pressure	direct/derived	indexed
TyresRimTemp	50	434274	C	tyre_temp_rim	direct/derived	indexed
TyresRubberTemp	10	86855	C	tyre_temp_rubber	direct/derived	indexed
TyresTempCentre	100	868548	C	tyre_temp, tyre_temp_centre	direct/derived	indexed
TyresTempLeft	100	868548	C	tyre_temp_left	direct/derived	indexed
TyresTempRight	100	868548	C	tyre_temp_right	direct/derived	indexed
Virtual Energy	20	173710	%	virtual_energy	direct/derived	indexed
Wheel Speed	100	868548	m/s	wheel_speed	direct/derived	indexed
WheelsDetached	-	1	-	wheels_detached	direct/derived	indexed
Wind Heading	1	8686	deg	wind_heading	direct/derived	indexed
Wind Speed	1	8686	m/s	wind_speed	direct/derived	indexed
Yellow Flag State	-	1	-	yellow_flag_state	direct/derived	indexed
channelsList	-	56	-	-	not used	indexed
eventsList	-	42	-	-	not used	indexed
metadata	-	11	-	-	not used	indexed

Appendix B-F - Events, Definitions & Limits

Event log

Lap 18 - pit transit [information]: entry-to-exit 115.0 s (high confidence)

Lap 42 - pit transit [information]: entry-to-exit 66.79999999999927 s (high confidence)

Lap 1 - observed impact signal [information]: magnitude 1.0 (medium confidence)

Lap 7 - off-track surface [warning]: 1 sampled contacts (high confidence)

Lap 7 - observed impact signal [information]: magnitude 1.0 (medium confidence)

Lap 13 - off-track surface [warning]: 1 sampled contacts (high confidence)

Lap 17 - off-track surface [warning]: 7 sampled contacts (high confidence)

Lap 19 - off-track surface [warning]: 3 sampled contacts (high confidence)

Lap 28 - off-track surface [warning]: 1 sampled contacts (high confidence)

Lap 31 - off-track surface [warning]: 1 sampled contacts (high confidence)

Event log continues on the next appendix page.

Appendix B-F - Events, Definitions & Limits

Event log

Lap 35 - off-track surface [warning]: 1 sampled contacts (high confidence)

Lap 44 - observed impact signal [information]: magnitude 1.0 (medium confidence)

Lap 52 - off-track surface [warning]: 1 sampled contacts (high confidence)

Derived metric definitions

- Representative pace: median of valid laps after pit, incomplete, flag, anomaly and shared quality exclusions.

- Fuel effect: Theil-Sen pairwise slope using at least six laps with fuel and pace; empirical, not a physical mass model.

- Tyre degradation: within-stint clean-pace slope where at least five comparable laps exist; correlated effects are disclosed.

- Theoretical best: sum of best valid sectors only when compatible sector splits exist; realistic potential remains separate.

- Traffic severity: derived from Time Behind Next when present; never treated as proof of an overtake.

Confidence and limitations

High: directly measured or strongly supported. Medium: consistent multi-channel inference. Low: plausible with missing context. Unavailable: insufficient source evidence. Exact overtakes, setup changes, opponent strategy, tyre/brake target windows, aerodynamic coefficients, reaction time, damage and race-control state are not fabricated.

Prioritized Actions

PRIORITY 1 - Review pit execution

Evidence: 2 timed pit transits were detected. Action: Compare entry-to-exit phases and rehearse the slowest repeatable phase. Expected benefit: Potential benefit cannot be isolated without stationary timing. Confidence: medium Validation: Record stationary phase and pit-cycle position state in the next race.

PRIORITY 2 - Stabilize representative race pace

Evidence: Robust lap-time MAD is 0.630 s. Action: Use clean-air laps as the pace baseline and flag traffic/pit laps separately. Expected benefit: A clearer strategy baseline; numerical gain is not supportable from current data. Confidence: high Validation: Compare the next race stint at similar fuel and tyre age.